

HK IMO Training 2020-2021
Problem Set 4
For 14 November 2020

Problem 13. Let a, b, c and d be real numbers with $a^2 + b^2 + c^2 + d^2 = 4$. Prove the inequality

$$(a + 2)(b + 2) \geq cd,$$

and find all equality cases.

Problem 14. Let $c > 1$ be a real number. Let $\{x_n\}$ be a sequence of real numbers such that $x_n > 1$ and

$$x_1 + x_2 + \cdots + x_n < cx_{n+1}$$

for all positive integer n . Prove that there exists a real number $a > 1$ such that $x_n > a^n$ for any positive integer n .

Problem 15. Given positive real numbers $a_1, a_2, \dots, a_n$, let

$$m = \min \left\{ a_1 + \frac{1}{a_2}, a_2 + \frac{1}{a_3}, \dots, a_{n-1} + \frac{1}{a_n}, a_n + \frac{1}{a_1} \right\}.$$

Prove the inequality

$$\sqrt[n]{a_1 a_2 \cdots a_n} + \frac{1}{\sqrt[n]{a_1 a_2 \cdots a_n}} \geq m.$$

Problem 16. Let a, b and c be positive real numbers. Find the minimum value of

$$\frac{(a + b)(b + c)(a + b + c)}{abc}.$$